Kazi Ababil Azam

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Research Interests

Intelligent Transportation Systems, Autonomous Vehicles, Cyber Physical Systems

Work Experience

Synesis IT PLC, Software Engineer *Software Development, React, WebRTC, TypeScript* Dhaka, Bangladesh
Jun 2024 to present

Contributed to the front-end development (React) of **Convay**, a video conferencing web application

Analyzed WebRTC statistics to monitor the performance of meetings in real-time, identifying areas for optimization

ERA InfoTech Limited, Project Intern *Natural Language Processing, Data Analysis* Dhaka, Bangladesh
May 2023 to Jun 2023

Worked on an internship project focused on real-life business data analysis through unsupervised clustering by applying natural language processing techniques

Education

B.Sc., **Bangladesh University of Engineering & Technology** Apr 2019 to Jul 2024
Department of Computer Science & Engineering (CSE) *CGPA: 3.67/4.00*

Notable Courses: Machine Learning, Microprocessor, Microcontroller & Embedded Systems, Compiler, Computer Networking, Software Engineering, Computer Architecture, Operating System, Simulation & Modelling, Computer Security, Computer Graphics, Artificial Intelligence

Research Experience

Towards Intelligent Traffic Signaling in Dhaka City Based on Vehicle Detection and Congestion Optimization Jun 2023 to Jun 2024

Supervised by: Dr. A. B. M. Alim Al Islam, Tarik Reza Toha, Masfiqur Rahaman

Development of a system architecture for traffic signal scheduling that incorporates successive implementations of vehicle detection and optimization algorithm in limited-resource environments, focused on the traffic scenario of Dhaka city.

Evaluation Metrics for Generative Pedestrian Crossing Models in Simulation May 2023 to present

Supervised by: Golam Md Muktadir, Dr. A. B. M. Alim Al Islam, Dr. Mahmuda Naznin

Analysis of real-life pedestrian behavior data from datasets in order to formulate metrics to compare and evaluate generative pedestrian models in autonomous vehicle simulations.

NHT1071: A Traffic Image Dataset for Non lane-based and Heterogeneous Traffic for Adaptive Traffic Signal Scheduling Jun 2023 to May 2024

Supervised by: Dr. A. B. M. Alim Al Islam, Masfiqur Rahaman

Collection and preparation of a dataset on traffic data at different intersections of Dhaka city for the purpose of developing intelligent traffic signaling systems suitable for the traffic in Dhaka city.

Academic Projects

SyncInc *Django, React, MUI, PostgreSQL* Github

A web-based project and task delegation management software designed for organizations. Stack includes PostgreSQL as the database, Django for the backend, ReactJS with Material UI for the frontend, Django REST framework for API development, and Django Channels for real-time notifications.

Laser Security System With Arduino *Arduino, Cyber Physical Systems* Github | YouTube

A project to control a laser security system with microcontrollers, where the ATMega32 unit manages the laser and the Arduino unit controls the alarm system and overall coordination.

Ray Tracing and Illumination Using OpenGL *C++, OpenGL* Github

Implementation of an image generation pipeline using ray tracing and illumination techniques to create realistic images of geometric shapes (sphere, pyramid, cube) and a 2D plane using OpenGL.

Implementation of a Congestion Control Algorithm for TCP Flows in ns-2 *ns-2* Github

Implementation of a research paper on congestion control in ns-2 codebase, specifically in TCP to achieve improved throughput in wired and wireless networks.

Compiler *Bison, Flex* Github

Built a compiler from scratch including steps for creating a symbol table, building a lexical analyzer, semantic analyzer, and generating intermediate code.

Technical Skills

Languages: C/C++, Python, Java, JavaScript, x86 Assembly, MySQL, Bison, Flex, Bash, Go, TypeScript

Frameworks: React, Bootstrap, Django, Express.js, Next.js

Technologies: LaTeX, Docker, Oracle DBMS, PostgreSQL, Git, Wireshark, Termius, Postman, OpenGL, ns-2

Libraries: OpenCV, PyAV, PyTorch, Keras, Pandas, Scikit-learn, SciPy, NLTK, Matplotlib, Seaborn, p5.js

Competitions

5th Place, IEEE CS, BUET presents "GameJam 2023" *p5.js* Github

Jan 2023

Secured the 5th position (out of 15 teams in the final round, 68 teams overall) by developing "Asteroids", a short, level based game using p5.js.

Leadership

Director of Events, BUET Cyber Security Club Link

Jun 2023 to Apr 2024

Promotion and organization of cybersecurity related events, seminars and both intra and inter-university Capture The Flag (CTF) competitions multiple times throughout the tenure

Test Scores

TOEFL: 113/120 (Reading: 29/30, Listening: 30/30, Speaking: 27/30, Writing: 27/30)